

Synthea EHR Analysis

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Data Loading

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#Load in Data
```

```
ehr_data <- read_csv("final_readmission_cohort.csv")
```

```
## Rows: 103867 Columns: 12
## -- Column specification -----
## Delimiter: ","
## chr  (6): patient_id, encounter_id, primary_diagnosis, gender, city, county
## dbl  (5): age_at_admission, length_of_stay, prior_admissions, is_30d_readmis...
## date (1): admit_date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#double check data load
```

```
glimpse(ehr_data)
```

```
## Rows: 103,867
## Columns: 12
## $ patient_id      <chr> "0000b92e-1559-9e6c-343d-babb57437157", "0000b92e-1~
## $ encounter_id     <chr> "0000b92e-1559-9e6c-f8e1-1621c684c56e", "0000b92e-1~
## $ primary_diagnosis <chr> "Limited social contact (finding)", "Concussion inj~
## $ age_at_admission <dbl> 20, 42, 42, 45, 12, 12, 36, 55, 55, 55, 55, 55,~
## $ gender           <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", "~
## $ city             <chr> "Lakeville", "Lakeville", "Lakeville", "Lakeville",~
```

```
## $ county          <chr> "Dakota County", "Dakota County", "Dakota County", ~
## $ admit_date      <date> 1997-04-16, 2019-07-24, 2019-07-24, 2022-08-03, 20~
## $ length_of_stay  <dbl> 0, 0, 0, 0, 0, 0, 7, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ prior_admissions <dbl> 0, 1, 1, 2, 0, 0, 0, 1, 1, 2, 2, 3, 4, 5, 6, 7, 8, ~
## $ is_30d_readmission <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ procedure_count  <dbl> 0, 0, 0, 2, 2, 2, 0, 5, 5, 6, 6, 0, 5, 0, 2, 0, 7, ~
```

Convert Variables to Factors and Create a Summary Table

```
#Convert binary and categorical variables into factors
```

```
ehr_clean <- ehr_data %>%
```

```
  mutate(
    is_30d_readmission = as.factor(is_30d_readmission),
    GENDER = as.factor(gender),
    COUNTY = as.factor(county),
    CITY = as.factor(city),
    DIAGNOSIS = as.factor(primary_diagnosis)
  )
```

```
#Summary of data with variables as factor
```

```
summary(ehr_clean)
```

```
##      patient_id      encounter_id      primary_diagnosis      age_at_admission
## Length :103867      Length :103867      Length :103867      Min. : 0.00
## N.unique : 9421      N.unique : 73406      N.unique : 162      1st Qu.: 41.00
## N.blank : 0          N.blank : 0          N.blank : 0          Median : 60.00
## Min.nchar: 36        Min.nchar: 36        Min.nchar: 4         Mean : 57.02
## Max.nchar: 36        Max.nchar: 36        Max.nchar: 78        3rd Qu.: 76.00
##                                     Max. :110.00
##
##      gender      city      county      admit_date
## Length :103867      Length :103867      Length :103867      Min. :1917-05-29
## N.unique : 2          N.unique : 1139      N.unique : 87      1st Qu.:2010-05-15
## N.blank : 0          N.blank : 0          N.blank : 0          Median :2018-12-30
## Min.nchar: 1          Min.nchar: 3          Min.nchar: 11       Mean :2014-04-12
## Max.nchar: 1          Max.nchar: 21         Max.nchar: 24       3rd Qu.:2022-10-23
##                                     Max. :2026-08-20
##
##      length_of_stay      prior_admissions      is_30d_readmission      procedure_count
## Min. : 0.0000      Min. : 0.00      0:58702      Min. : 0.000
## 1st Qu.: 0.0000      1st Qu.: 2.00      1:45165      1st Qu.: 0.000
## Median : 0.0000      Median : 10.00                      Median : 3.000
## Mean : 0.6777      Mean : 35.43                      Mean : 3.826
## 3rd Qu.: 0.0000      3rd Qu.: 42.00                      3rd Qu.: 6.000
## Max. :153.0000      Max. :507.00                      Max. :89.000
##
##      GENDER      COUNTY      CITY
## F:45175      Hennepin County :21709      Minneapolis : 8842
## M:58692      Ramsey County :11585      St. Paul : 7937
##              Dakota County : 8369      Plymouth : 1849
##              Anoka County : 7699      Apple Valley: 1811
##              Washington County: 4579      Rochester : 1755
```

```
##           St. Louis County : 3290   Lakeville   : 1689
##           (Other)          :46636   (Other)     :79984
##           DIAGNOSIS
## NULL                       :25785
## Medication review due (situation):18082
## Stress (finding)           : 6444
## Full-time employment (finding) : 5520
## Gingivitis (disorder)       : 5276
## Part-time employment (finding) : 3951
## (Other)                    :38809
```

Summary Table with Gender and Demography

```
## Summary Table with Gender and Demography

library(gtsummary)

#Build the stratified summary table
ehr_clean %>%
  #Select the specific columns in the summary
  select(age_at_admission, GENDER, COUNTY, procedure_count, length_of_stay, is_30d_readmission) %>%
  #Group by if they were readmitted
  tbl_summary(by = is_30d_readmission) %>%
  #Add p-values to compare
  add_p()
```

```
## The following errors were returned during 'add_p()':
## x For variable 'COUNTY' ('is_30d_readmission') and "estimate", "p.value",
## "conf.low", and "conf.high" statistics: FEXACT error 40. Out of workspace.
```

Top 15 Clinical Drivers of Readmission Graph

```
library(ggplot2)
library(stringr)

#Top 15 Clinical Drivers of Readmission
ehr_clean %>%
  #Step 1: Filter for patients who were readmitted
  filter(is_30d_readmission == "1") %>%

  #Step 2: Remove missing data and NULL
  drop_na(DIAGNOSIS) %>%
  filter(DIAGNOSIS != "NULL") %>%

  #Step 3: Filter out Social Health determinants and administrative noise
  filter(!str_detect(str_to_lower(DIAGNOSIS), "review|employment|labor|social|violence|finding|situation"))

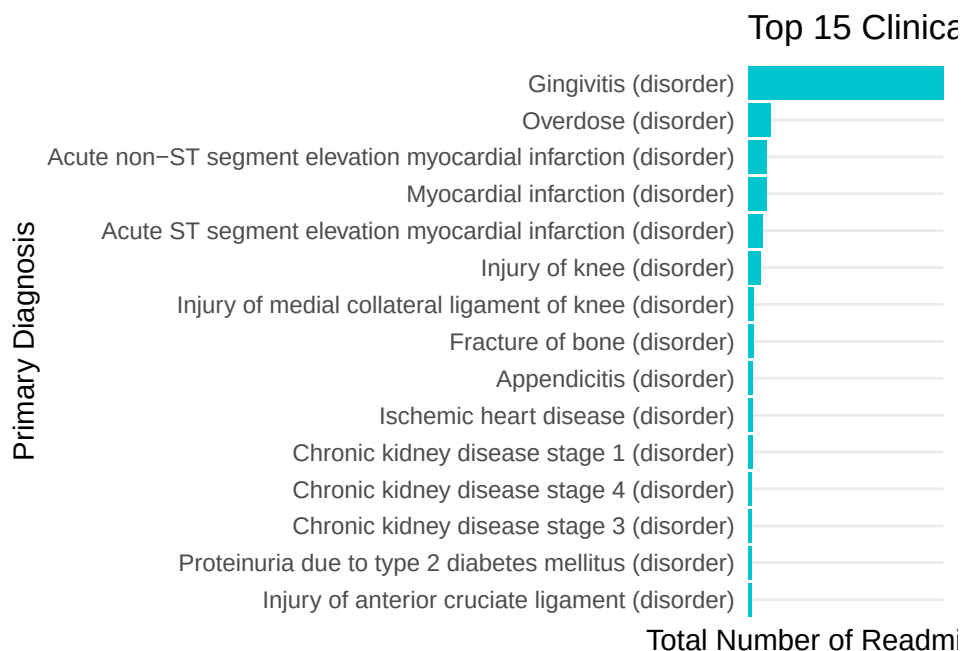
  #Step 4: Count the number of occurrences of each specific diagnosis
  count(DIAGNOSIS, name = "readmission_count") %>%
```

```

#Step 5: Keep only the top 15 most frequent conditions
slice_max(order_by = readmission_count, n = 15) %>%

#Step 6: Build the plot
ggplot(aes(x = reorder(DIAGNOSIS, readmission_count), y = readmission_count)) +
  geom_col(fill = "turquoise3") +
  coord_flip() +
  labs(
    title = "Top 15 Clinical Drivers of 30-Day Readmissions",
    x = "Primary Diagnosis",
    y = "Total Number of Readmissions"
  ) +
  scale_y_discrete(labels = \(x) str_wrap(x, width = 30)) +
  theme_minimal()

```



Top 15 Social Drivers of Readmission Graph

```

#The Social Determinants Plot: Non-Clinical Drivers of Readmission
ehr_clean %>%

#Step 1: Filter for patients who were readmitted
filter(is_30d_readmission == "1") %>%

#Step 2: Remove missing data and NULL
drop_na(DIAGNOSIS) %>%
filter(DIAGNOSIS != "NULL") %>%

#Step 3: Drop the medication review (non clinical and not readmission)
filter(!str_detect(str_to_lower(DIAGNOSIS), "medication review")) %>%

```

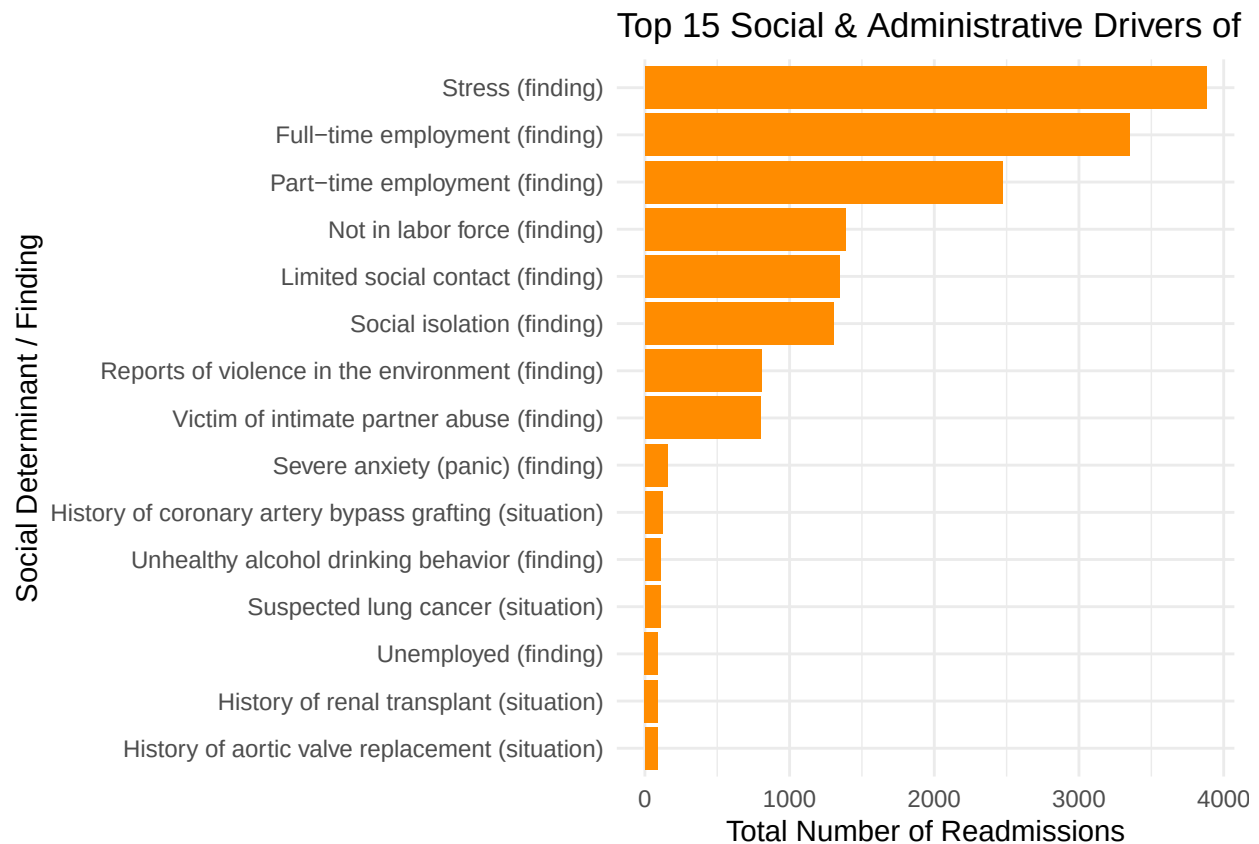
```

#Step 4: Filter the social and administrative findings
filter(str_detect(str_to_lower(DIAGNOSIS), "employment|labor|social|stress|finding|situation")) %>%

#Step 5: Count and slice the top 15
count(DIAGNOSIS, name = "readmission_count") %>%
slice_max(order_by = readmission_count, n = 15) %>%

#Step 6: Build the plot
ggplot(aes(x = reorder(DIAGNOSIS, readmission_count), y = readmission_count)) +
geom_col(fill = "darkorange") +
coord_flip() +
labs(
  title = "Top 15 Social & Administrative Drivers of 30-Day Readmissions",
  x = "Social Determinant / Finding",
  y = "Total Number of Readmissions"
) +
theme_minimal()

```



Highest Readmission Rate by County Graph

```

#Load the scales package to format our axis as percentages
library(scales)

```

```

##
## Attaching package: 'scales'

```

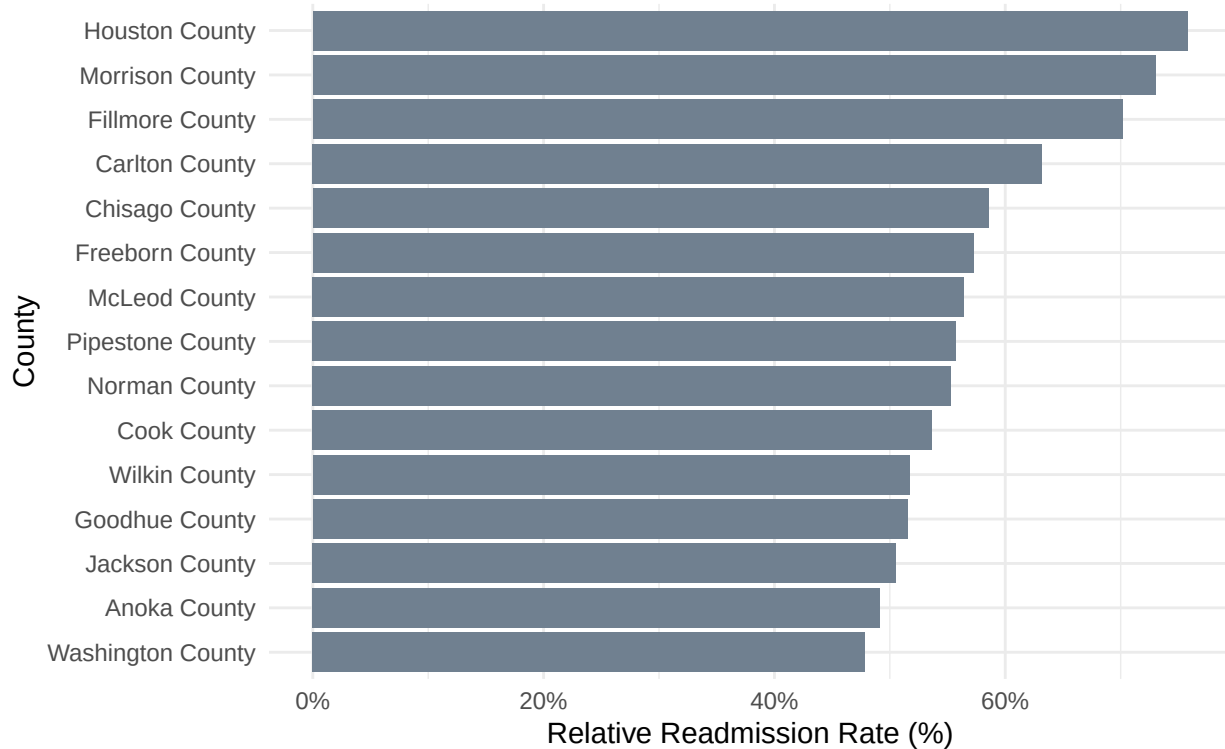
```
## The following object is masked from 'package:purrr':  
##  
##      discard
```

```
## The following object is masked from 'package:readr':  
##  
##      col_factor
```

```
ehr_clean %>%  
  #Step 1: Remove missing county data  
  drop_na(COUNTY) %>%  
  
  #Step 2: Group the data by County  
  group_by(COUNTY) %>%  
  
  #Step 3: Calculate totals and relative frequencies (percentages)  
  summarize(  
    total_visits = n(),  
    readmitted_count = sum(is_30d_readmission == "1"),  
    readmission_rate = readmitted_count / total_visits  
  ) %>%  
  
  #Step 4: The Sample Size Filter  
  #Drop any county with fewer than 30 total visits to eliminate statistical noise  
  filter(total_visits >= 30) %>%  
  
  #Step 5: Keep the top 15 counties with the highest readmission rates  
  slice_max(order_by = readmission_rate, n = 15) %>%  
  
  #Step 6: Build the plot  
  ggplot(aes(x = reorder(COUNTY, readmission_rate), y = readmission_rate)) +  
  geom_col(fill = "slategray") +  
  coord_flip() +  
  
  #Format the Y-axis (which is flipped to the bottom) as percentages  
  scale_y_continuous(labels = percent_format(accuracy = 1)) +  
  labs(  
    title = "Highest 30-Day Readmission Rates by County",  
    subtitle = "Filtered for counties with n >= 30 hospital visits",  
    x = "County",  
    y = "Relative Readmission Rate (%)"  
  ) +  
  theme_minimal()
```

Highest 30-Day Readmission Rates by County

Filtered for counties with n >= 30 hospital visits



Logistic Regression Model

```
library(broom)
```

```
#Step 1: Build the Logistic Regression Model
#Testing if readmission is driven by Age and Gender
readmission_model <- glm(
  is_30d_readmission ~ age_at_admission + GENDER,
  data = ehr_clean,
  family = "binomial" #This tells R it's a Yes/No (1 or 0) outcome
)
```

```
#Step 2: Tidy the results and calculate the Odds Ratios with CI
tidy(readmission_model, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 3 x 7
##   term                estimate std.error statistic  p.value  conf.low  conf.high
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)         0.0516  0.0233   -127.    0         0.0493   0.0540
## 2 age_at_admission    1.05    0.000354  125.    0         1.04     1.05
## 3 GENDER              1.17    0.0142    11.3 1.08e-29  1.14     1.21
```

Characteristic	0 N = 58,702 ¹	1 N = 45,165 ¹	p-value ²
age_at_admission	50 (28, 66)	70 (58, 81)	<0.001
GENDER			<0.001
F	28,230 (48%)	16,945 (38%)	
M	30,472 (52%)	28,220 (62%)	
COUNTY			
Aitkin County	165 (0.3%)	40 (<0.1%)	
Anoka County	3,914 (6.7%)	3,785 (8.4%)	
Becker County	309 (0.5%)	149 (0.3%)	
Beltrami County	630 (1.1%)	349 (0.8%)	
Benton County	271 (0.5%)	65 (0.1%)	
Big Stone County	60 (0.1%)	6 (<0.1%)	
Blue Earth County	798 (1.4%)	319 (0.7%)	
Brown County	236 (0.4%)	51 (0.1%)	
Carlton County	320 (0.5%)	549 (1.2%)	
Carver County	933 (1.6%)	577 (1.3%)	
Cass County	279 (0.5%)	219 (0.5%)	
Chippewa County	127 (0.2%)	33 (<0.1%)	
Chisago County	828 (1.4%)	1,169 (2.6%)	
Clay County	492 (0.8%)	259 (0.6%)	
Clearwater County	49 (<0.1%)	11 (<0.1%)	
Cook County	89 (0.2%)	103 (0.2%)	
Cottonwood County	119 (0.2%)	43 (<0.1%)	
Crow Wing County	654 (1.1%)	485 (1.1%)	
Dakota County	4,839 (8.2%)	3,530 (7.8%)	
Dodge County	183 (0.3%)	94 (0.2%)	
Douglas County	357 (0.6%)	253 (0.6%)	
Faribault County	89 (0.2%)	11 (<0.1%)	
Fillmore County	212 (0.4%)	499 (1.1%)	
Freeborn County	567 (1.0%)	759 (1.7%)	
Goodhue County	765 (1.3%)	814 (1.8%)	
Grant County	52 (<0.1%)	20 (<0.1%)	
Hennepin County	12,323 (21%)	9,386 (21%)	
Houston County	271 (0.5%)	849 (1.9%)	
Hubbard County	311 (0.5%)	160 (0.4%)	
Isanti County	335 (0.6%)	178 (0.4%)	
Itasca County	593 (1.0%)	328 (0.7%)	
Jackson County	189 (0.3%)	193 (0.4%)	
Kanabec County	219 (0.4%)	160 (0.4%)	
Kandiyohi County	398 (0.7%)	272 (0.6%)	
Kittson County	43 (<0.1%)	5 (<0.1%)	
Koochiching County	38 (<0.1%)	25 (<0.1%)	
Lac qui Parle County	149 (0.3%)	133 (0.3%)	
Lake County	138 (0.2%)	60 (0.1%)	
Lake of the Woods County	30 (<0.1%)	26 (<0.1%)	
Le Sueur County	264 (0.4%)	152 (0.3%)	
Lincoln County	25 (<0.1%)	20 (<0.1%)	
Lyon County	167 (0.3%)	7 (<0.1%)	
Mahnommen County	57 (<0.1%)	25 (<0.1%)	
Marshall County	128 (0.2%)	43 (<0.1%)	
Martin County	180 (0.3%)	41 (<0.1%)	